Flux, Spectrum, Density, and Anisotropic Boundary Conditions Used by AMPS

AMPS Interface Documentation

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Abstract

This note documents the quantities used by the AMPS gridless energetic-particle post-processing tools for cutoff rigidity, local spectrum, omnidirectional flux, and number density. It also describes how isotropic and anisotropic boundary spectra are implemented in the current code, and how the internal dipole-field reference solutions used for nightly tests are constructed for comparison with numerical model output in both `POINTS` and `SHELLS` modes.

1 Differential intensity and related quantities

Throughout the AMPS interface, the particle spectrum is specified as a *differential directional intensity*

$$J(E,\Omega) \equiv \frac{dN}{dA\,dt\,d\Omega\,dE}, \tag{1}$$

with typical units

$$J(E,\Omega)\ [\text{particle m}^{-2}\,\text{s}^{-1}\,\text{sr}^{-1}\,\text{MeV}^{-1}]. \tag{2}$$

The boundary spectrum supplied to the gridless density/spectrum tool is a model for this quantity at the outer boundary of the tracing domain.

Energy variable. For ions, AMPS often uses kinetic energy per nucleon in MeV/nucleon in the user interface. Internally, the density/spectrum integration is carried out over an energy grid E_i in MeV; the same spectral model is queried at each energy point and folded with geomagnetic transmissivity.

Omnidirectional differential flux. The omnidirectional differential flux is the solid-angle integral of the intensity,

$$j(E) = \int_{4\pi} J(E,\Omega)\,d\Omega. \tag{3}$$

If the distribution is isotropic, then

$$j(E) = 4\pi J(E). \tag{4}$$

In anisotropic cases the factor 4π cannot be inserted by hand; the angular integral must be evaluated with the actual angular weighting.

Integral flux and pfu. The omnidirectional integral flux above threshold E_0 is

$$F(> E_0) = \int_{E_0}^{\infty} j(E)\, dE, \tag{5}$$

or, equivalently, in isotropic form,

$$F(> E_0) = 4\pi \int_{E_0}^{\infty} J(E)\, dE. \tag{6}$$

NOAA/SWPC often reports proton event fluxes in pfu, where

$$1 \text{ pfu} = 1 \text{ proton cm}^{-2}\,\text{s}^{-1}\,\text{sr}^{-1} \tag{7}$$

for an *integral* channel. When fitting a model to integral channels, AMPS computes the energy integral of the chosen spectral model.

2 From intensity to local spectrum, flux, and density

The gridless solver does not solve a full transport equation in phase space. Instead, for each evaluation location $\boldsymbol{x}_0$ and energy E, it computes the fraction of directions at the outer boundary that can access that point by backtracing particles in the geomagnetic field. This fraction is the *transmissivity* $T(E, \boldsymbol{x}_0)$.

2.1 Boundary spectrum and local spectrum

Let $J_b(E)$ denote the boundary differential intensity model at the outer boundary. In the isotropic case the local differential intensity is

$$J_{\text{loc}}(E; \boldsymbol{x}_0) = T(E, \boldsymbol{x}_0)\, J_b(E). \tag{8}$$

Thus the magnetosphere acts as an energy- and location-dependent filter on the specified boundary spectrum.

2.2 Number density from intensity

For an isotropic distribution, the differential number density $n_E(E)$ and intensity are related by

$$n_E(E) = \frac{4\pi}{v(E)} J(E), \tag{9}$$

where $v(E)$ is the particle speed corresponding to kinetic energy E. The number density is therefore

$$n(\boldsymbol{x}_0) = 4\pi \int \frac{J_{\text{loc}}(E; \boldsymbol{x}_0)}{v(E)}\, dE. \tag{10}$$

This factor $1/v(E)$ is physically important: low-energy particles contribute more strongly to density than to integral flux because they linger longer per unit path length.

2.3 Omnidirectional integral flux

The total omnidirectional integral flux over the modeled energy band is

$$F(\boldsymbol{x}_0) = 4\pi \int J_{\mathrm{loc}}(E; \boldsymbol{x}_0)\, dE. \tag{11}$$

For a user-defined channel $[E_1, E_2]$, the channel flux is

$$F_{[E_1,E_2]}(\boldsymbol{x}_0) = 4\pi \int_{E_1}^{E_2} J_{\mathrm{loc}}(E; \boldsymbol{x}_0)\, dE. \tag{12}$$

The code evaluates Eqs. (10)–(12) with trapezoidal quadrature on the same energy grid used for transmissivity.

2.4 Relativistic conversion between energy and speed

Let m_0 be the particle rest mass and T the kinetic energy in joules. The Lorentz factor is

$$\gamma = 1 + \frac{T}{m_0 c^2}, \tag{13}$$

and the speed is

$$v(E) = c\sqrt{1 - \frac{1}{\gamma^2}}. \tag{14}$$

This relativistic speed is used consistently when converting local intensity to number density.

3 Isotropic and anisotropic boundary spectra

3.1 Isotropic boundary mode

If the boundary distribution is isotropic, the boundary intensity depends only on energy,

$$J_b(E, \Omega) = J_b(E). \tag{15}$$

In that case the only angular dependence entering the local solution is the geomagnetic accessibility. The code samples a finite direction set $\{\Omega_k\}_{k=1}^{N_\Omega}$ and computes

$$T(E, \boldsymbol{x}_0) \approx \frac{1}{N_\Omega} \sum_{k=1}^{N_\Omega} I_k(E, \boldsymbol{x}_0), \tag{16}$$

where $I_k = 1$ if the reversed trajectory launched from $\boldsymbol{x}_0$ in direction Ω_k escapes the outer boundary and $I_k = 0$ otherwise.

Substituting Eq. (16) into Eq. (8) yields the discrete isotropic approximation implemented in the code.

3.2 Anisotropic boundary mode

If the boundary distribution is anisotropic, the code models it as an isotropic spectral shape multiplied by a directional and/or spatial weighting factor. In that case the boundary intensity is written schematically as

$$J_b(E, \Omega, \boldsymbol{x}_{\text{exit}}) = J_b(E)\, f_{\text{aniso}}(\Omega, \boldsymbol{x}_{\text{exit}}), \tag{17}$$

where $\boldsymbol{x}_{\text{exit}}$ is the position where the backtraced trajectory reaches the outer boundary. In the present implementation the anisotropy factor is separated into

$$f_{\text{aniso}} = f_{\text{PAD}}(\mu)\, f_{\text{spatial}}(\boldsymbol{x}_{\text{exit}}), \qquad \mu = \cos\alpha, \tag{18}$$

where α is the pitch angle with respect to the local magnetic field at the exit point.

The code therefore computes a weighted transmissivity

$$T_{\text{aniso}}(E, \boldsymbol{x}_0) \approx \frac{\sum_{k=1}^{N_\Omega} I_k(E, \boldsymbol{x}_0)\, w_k}{\sum_{k=1}^{N_\Omega} w_k}, \qquad w_k = f_{\text{PAD}}(\mu_k)\, f_{\text{spatial}}(\boldsymbol{x}_{\text{exit},k}). \tag{19}$$

If all weights are unity, Eq. (19) reduces to the isotropic average in Eq. (16).

3.3 Physics of the anisotropy model

The anisotropic option is meant to represent the physically common situation in which the outer-boundary population is not uniformly distributed in direction or space. Examples include:

- field-aligned or anti-field-aligned streaming following solar injection,
- bidirectional populations produced by magnetic mirroring or trapping,
- day-night or regional asymmetries in the outer magnetosphere, and
- user-prescribed pitch-angle distributions for sensitivity studies.

The purpose of the implementation is not to solve a self-consistent kinetic transport problem at the boundary, but to apply a physically transparent weighting model to the trajectories that successfully connect the outer boundary to the observation point.

3.4 Pitch-angle distribution models

The code supports several pitch-angle weighting laws through `EvalAnisotropyFactor`:

- `ISOTROPIC`: $f_{\text{PAD}} = 1$;
- `SINALPHA_N`: $f_{\text{PAD}} = (\sin\alpha)^N = (1-\mu^2)^{N/2}$;
- `COSALPHA_N`: $f_{\text{PAD}} = |\cos\alpha|^N = |\mu|^N$;
- `BIDIRECTIONAL`: a symmetric enhancement about $\mu = \pm 1$.

These provide a simple way to emulate broad classes of distributions ranging from quasi-perpendicular populations to strongly field-aligned beams.

3.5 Spatial weighting models

The present implementation also permits a multiplicative spatial weighting, $f_{\text{spatial}}(\boldsymbol{x}_{\text{exit}})$, at the exit point of the backtraced trajectory. For example, a day-night prescription can assign different weights on the dayside and nightside outer boundary. The physical interpretation is that the intensity incident on the outer boundary is not spatially uniform, so trajectories connecting to favored boundary regions contribute more strongly to the local population.

3.6 Local spectrum, density, and flux in anisotropic mode

Once the weighted transmissivity of Eq. (19) is obtained, the remaining formulas are unchanged:

$$J_{\mathrm{loc}}(E;\boldsymbol{x}_0) = T_{\mathrm{aniso}}(E,\boldsymbol{x}_0)\, J_b(E), \tag{20}$$

$$n(\boldsymbol{x}_0) = 4\pi \int \frac{J_{\mathrm{loc}}(E;\boldsymbol{x}_0)}{v(E)}\, dE, \tag{21}$$

$$F(\boldsymbol{x}_0) = 4\pi \int J_{\mathrm{loc}}(E;\boldsymbol{x}_0)\, dE. \tag{22}$$

Thus anisotropy changes the *effective transmission factor*, not the fundamental definitions of spectrum, density, or flux.

4 Numerical algorithm implemented in the gridless solver

For either `POINTS` or `SHELLS` mode, the density/spectrum solver follows the same algorithm.

1. Build the requested energy grid $\{E_i\}$, either linearly or logarithmically spaced.
2. Build the discrete direction set $\{\Omega_k\}$ used to sample arrival directions at the evaluation point.
3. For each location $\boldsymbol{x}_0$ and energy E_i, convert energy to rigidity R_i and backtrace reversed trajectories in the prescribed magnetic field.
4. For each sampled direction, determine whether the trajectory reaches the outer boundary (allowed) or intersects the inner loss sphere / fails the termination criterion (forbidden).
5. Form the transmissivity using either the unweighted isotropic average, Eq. (16), or the weighted anisotropic average, Eq. (19).
6. Evaluate the boundary spectrum $J_b(E_i)$ and form the local spectrum $J_{\mathrm{loc}}(E_i) = T(E_i)J_b(E_i)$.
7. Integrate the local spectrum over energy to obtain total number density, total omnidirectional integral flux, and any requested energy-channel fluxes.
8. Write output files containing the total scalar fields and, where requested, the full transmissivity and local spectrum as functions of energy.

POINTS mode. In `POINTS` mode the tool evaluates a user-supplied list of locations and writes pointwise files for density, spectrum, and flux.

SHELLS mode. In `SHELLS` mode the tool evaluates a structured longitude-latitude grid on one or more altitude shells and writes the corresponding shell maps. The local physics and energy integration are identical to `POINTS` mode; only the sampling of $\boldsymbol{x}_0$ differs.

5 Analytic and semi-analytic dipole reference solutions used for nightly tests

To validate the implementation, the code contains internal reference calculations for the special case of a dipole geomagnetic field. These references are written to comparison files such as

`density_gridless_points_dipole_compare.dat`,
`spectrum_gridless_points_dipole_compare.dat`,
`flux_gridless_points_dipole_compare.dat`

and their `SHELLS` analogs when nightly-test mode is enabled.

5.1 Purpose of the dipole benchmark

The benchmark is designed to test the entire chain

$$\text{cutoff} \to \text{transmissivity} \to \text{local spectrum} \to \text{density / flux integrals}, \tag{23}$$

while using a field model simple enough that the dominant geometric behavior is understood analytically. It is not intended to replace the full numerical model, but to provide a stable and interpretable regression test.

5.2 Vertical Stórmer cutoff used in the reference

For a dipole field, the classical Stórmer theory provides an analytic cutoff rigidity for special geometries. The internal benchmark uses the *vertical* Stórmer cutoff,

$$R_{c,\mathrm{vert}}(\lambda, r) \propto \frac{\cos^4 \lambda}{r^2}, \tag{24}$$

where λ is magnetic latitude and r is radial distance in Earth radii. The code evaluates this quantity through an internal helper and uses it as a hard transition in rigidity space:

$$H(E; \boldsymbol{x}_0) = \begin{cases} 0, & R(E) < R_{c,\mathrm{vert}}(\boldsymbol{x}_0), \\ 1, & R(E) \geq R_{c,\mathrm{vert}}(\boldsymbol{x}_0). \end{cases} \tag{25}$$

Equation (25) is the spectrum-space analogue of the usual analytic dipole cutoff benchmark.

5.3 Isotropic geometric factor in the reference

The numerical solver uses curved trajectories in the dipole field. The internal analytic reference intentionally uses a simpler geometry: a straight-line "open-sky" estimate for the fraction of directions from which a particle can reach the outer boundary without intersecting the inner spherical absorber. Let $G_{\mathrm{sky}}(\boldsymbol{x}_0)$ denote that purely geometric fraction of accessible solid angle. Then the isotropic reference transmissivity is

$$T_{\mathrm{ref}}^{\mathrm{iso}}(E, \boldsymbol{x}_0) = G_{\mathrm{sky}}(\boldsymbol{x}_0)\, H(E; \boldsymbol{x}_0). \tag{26}$$

This separates the problem into a geometric factor and an energy threshold.

5.4 Anisotropic geometric factor in the reference

For anisotropic nightly tests, the reference cannot be reduced to a single constant open-sky factor because different accessible directions carry different weights. The code therefore computes a *semi-analytic* reference by integrating the prescribed anisotropy weight over the straight-line accessible solid angle:

$$G_{\mathrm{sky}}^{\mathrm{aniso}}(\boldsymbol{x}_0) = \frac{\int_{\Omega \in \Omega_{\mathrm{open}}} f_{\mathrm{aniso}}(\Omega, \boldsymbol{x}_{\mathrm{exit}})\, d\Omega}{\int_{4\pi} f_{\mathrm{aniso}}(\Omega, \boldsymbol{x}_{\mathrm{exit}})\, d\Omega}, \tag{27}$$

where Ω_{open} is the subset of directions that do not intersect the inner absorber in the straight-line construction. The anisotropic reference transmissivity is then

$$T_{\mathrm{ref}}^{\mathrm{aniso}}(E, \boldsymbol{x}_0) = G_{\mathrm{sky}}^{\mathrm{aniso}}(\boldsymbol{x}_0)\, H(E; \boldsymbol{x}_0). \tag{28}$$

This benchmark is still much simpler than the full backtracing solution, but it is sensitive to errors in anisotropy weighting, angular normalization, and the subsequent spectrum-folding and integration steps.

5.5 Reference local spectrum, density, and flux

Once $T_{\mathrm{ref}}(E, \boldsymbol{x}_0)$ is known from either Eq. (26) or Eq. (28), the reference local spectrum is

$$J_{\mathrm{loc,ref}}(E; \boldsymbol{x}_0) = T_{\mathrm{ref}}(E, \boldsymbol{x}_0)\, J_b(E). \tag{29}$$

The reference density and flux are then computed by the *same* formulas as the numerical solution,

$$n_{\mathrm{ref}}(\boldsymbol{x}_0) = 4\pi \int \frac{J_{\mathrm{loc,ref}}(E; \boldsymbol{x}_0)}{v(E)}\, dE, \tag{30}$$

$$F_{\mathrm{ref}}(\boldsymbol{x}_0) = 4\pi \int J_{\mathrm{loc,ref}}(E; \boldsymbol{x}_0)\, dE, \tag{31}$$

$$F_{\mathrm{ref},[E_1,E_2]}(\boldsymbol{x}_0) = 4\pi \int_{E_1}^{E_2} J_{\mathrm{loc,ref}}(E; \boldsymbol{x}_0)\, dE. \tag{32}$$

Using the same quadrature and the same boundary-spectrum evaluator in both the reference and numerical branches isolates errors in geomagnetic filtering and anisotropy handling from errors in spectral integration.

5.6 What is compared in the output files

The comparison files report, depending on the product:

- numerical versus reference density,
- numerical versus reference transmissivity spectra,
- numerical versus reference local spectra, and
- numerical versus reference total and channel-integrated fluxes.

These outputs are intentionally separated because different bugs show up most clearly in different products:

- a cutoff or rigidity-conversion bug appears first in transmissivity and local spectrum,
- a speed-conversion bug appears most strongly in number density because of the factor $1/v(E)$, and
- a channel-integration or interpolation bug appears most strongly in the integrated flux outputs.

6 Rigidity spectra and variable changes

Some literature expresses spectra in rigidity R instead of energy:

$$J_R(R) = \frac{dN}{dA\, dt\, d\Omega\, dR}. \tag{33}$$

The conversion between energy-space and rigidity-space spectra is

$$J_R(R) = J_E(E) \left| \frac{dE}{dR} \right|, \tag{34}$$

where the mapping $E(R)$ depends on particle mass and charge because $R = pc/(Ze)$. The cutoff-rigidity solver and the density/spectrum solver therefore share the same species-dependent energy-rigidity conversion.

7 Implementation notes for the current AMPS code

The present gridless density/spectrum tool uses the same trajectory classifier as the gridless cutoff-rigidity tool. This is important for consistency: the density solver does not maintain a separate magnetic-field evaluator, particle pusher, or termination logic. Instead, both cutoff and density calculations rely on the same single-particle accessibility test, and differ only in how they post-process the result:

- cutoff rigidity seeks the threshold in rigidity for a given location and direction set;
- density/spectrum computes transmissivity over an energy grid and folds it with a boundary spectrum.

This shared-core design reduces the risk that the two tools drift apart because of subtle implementation differences.

8 Summary

The AMPS gridless density/spectrum model computes local energetic-particle properties by filtering a specified boundary spectrum through geomagnetic transmissivity. In isotropic mode, transmissivity is the unweighted fraction of allowed directions; in anisotropic mode it is a weighted fraction reflecting the prescribed pitch-angle and spatial dependence at the outer boundary. The local spectrum is then integrated over energy to obtain omnidirectional flux and number density. For the dipole field, AMPS includes built-in analytic and semi-analytic reference solutions based on the vertical Stórmer cutoff and a straight-line open-sky geometry, making it possible to compare numerical and reference results for density, spectrum, and flux in both `POINTS` and `SHELLS` modes without relying on external scripts.

9 References (selected)

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